



C23-EC/ECII/CIOT-104

23015

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

FIRST YEAR (COMMON) EXAMINATION

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. Write charge and mass of electron, proton and neutron.
- 2. Write the electronic configuration of carbon, sodium and chlorine.
- 3. Define buffer solution. Write any two applications of it.
- 4. Define solute, solvent and solution.
- 5. Write any three factors which influence the rate of corrosion.
- 6. Write any three differences between electrolytic conduction and metallic conduction.
- 7. Write any three disadvantages of using hard water in industries.
- 8. Define elastomer. Give any two examples.

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9. Write the composition and two uses of CNG.

10. Write a short note in 'Acid rain'.

PART—B

10×5=50

Instructions : (1) Answer *any five* questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. State the postulates of Bohr's atomic theory. Write its drawbacks. 8+2

12. (a) Define Normality. Find the Normality of 3.65 gms of HCl dissolved in 500 ml solution. 6

(b) Write any four differences between properties of ionic and covalent compounds. 4

13. (a) Define Ionic bond. Explain the formation of Ionic bond in NaCl. 5

(b) Explain Arrhenius theory of acid and bases with suitable examples. 5

14. (a) Explain working of lead storage battery. 5

(b) Explain the electrolysis of fused NaCl. 5

15. (a) Explain the prevention of corrosion of metals by sacrificial anode method. 5

(b) Write short notes on stress cell and concentration cell. 5

- 16.** (a) A water sample contains following substances. 6
- (i) $\text{Ca}(\text{HCO}_3)_2$ – 8.1 mg/lit (ii) MgCl_2 – 19 mg/lit
- (iii) CaSO_4 – 6.8 mg/lit (iv) MgSO_4 – 12 mg/lit
- Calculate the total hardness of water in terms of CaCO_3 equivalents.
- (b) Define reverse osmosis. Write any two advantages of it. 4
- 17.** Write the preparation and two uses of (a) PVC (b) Teflon (c) Polystyrene and (d) Nylon-66. 10
- 18.** (a) Define Deforestation. State any four causes of deforestation. 4
- (b) What are renewable and non-renewable energy sources? Give two examples for each. 6

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